

6 (**short**) A college is considering whether to heat a building with a gas boiler or an electrically-driven heat pump.

(a) If the heat pump operates between a cold-side temperature T_c and a hot-side temperature T_h , what is its maximum possible coefficient of performance. [2]

(b) Due to irreversibilities in the heat pump, the actual work required is a multiple α of the work that would be required if the heat pump were reversible. What is the actual coefficient of performance of the heat pump? [2]

(c) If the electricity is generated exclusively by a gas-fired power station that converts a proportion, η , of the chemical power into electrical power, at what ratio T_h/T_c would the gas boiler be more energy-efficient than the heat pump? For simplicity, you may assume that the gas boiler converts all of the chemical power into heat. [2]

(d) If $T_c = 273$ K, $\eta = 0.6$, and $\alpha = 2.0$, at what value of T_h would consumption of gas be reduced by using the gas boiler instead of the heat pump? Comment on your answer. What other engineering factors should the college consider? [4]